CrowdMAPS: Crowdsourcing Multiple Actions per State for Real-World Robot Imitation Learning

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The Hardware Bottleneck

The crowd can scale. Robot data crowdsourcing remains hardware-bound.

- Prior crowdsourcing systems pairs **one operator to one robot at one time**.
- When contributors outnumber robots, **people wait**.
- **Hardware still dictates throughput.**

Missing Supervision?

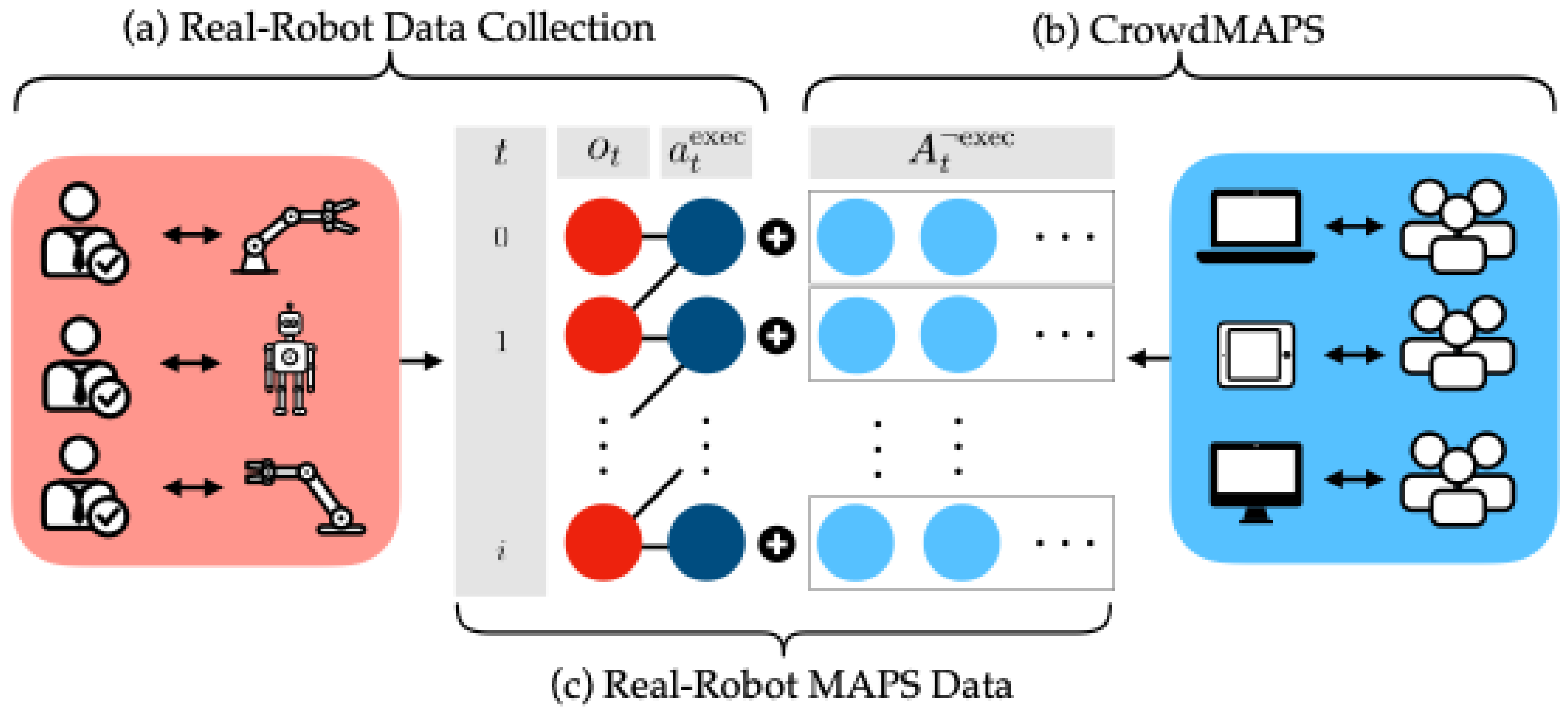
One state. Many valid actions. But only one recorded.

- Standard demonstrations capture **only the executed action**, leaving other valid actions unobserved.

What if we crowdsource more actions from the states we've already collected, instead of only collecting more trajectories?

Action Densification: A New Scaling Axis

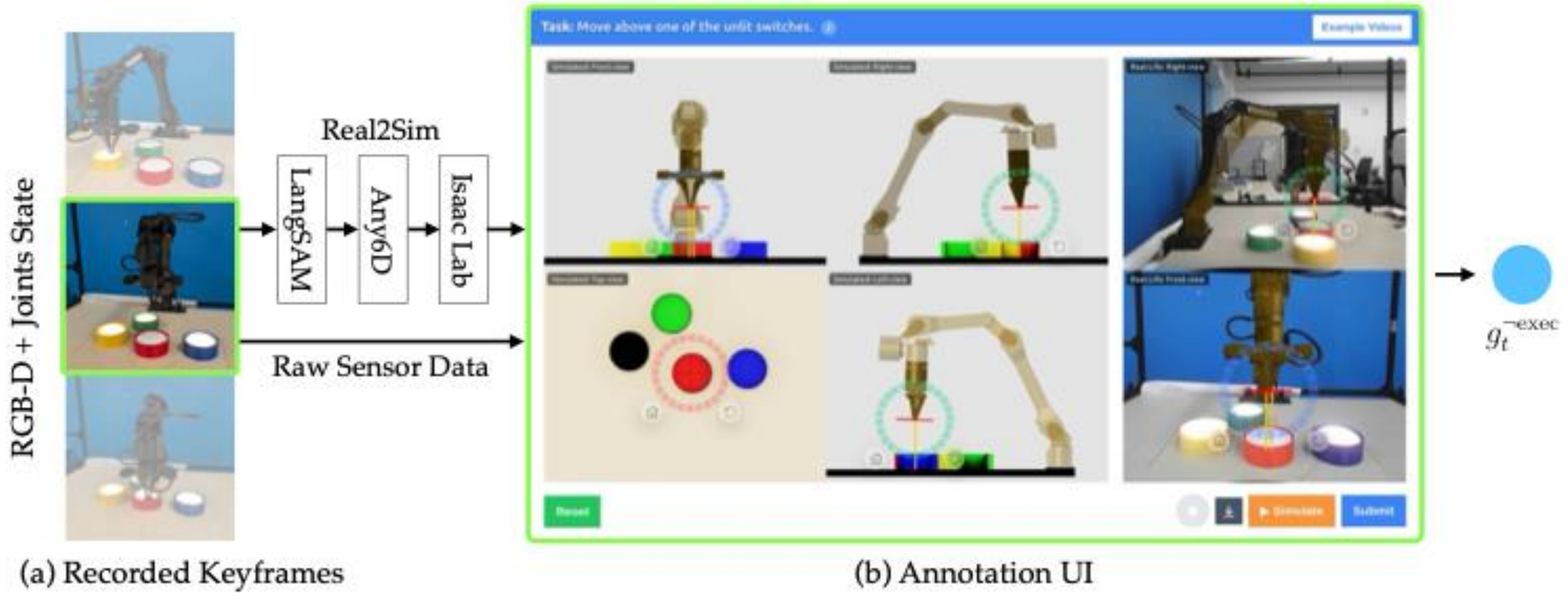
Scale robot data in width, not just length.
Length: More trajectories, more physically visited states.
Width: More valid actions for states already visited.
MAPS densifies action supervision without adding physical state transitions.



Decoupling Action Annotation from Robot Hardware

Physical data collection acquires states. The crowd provides alternative action labels.

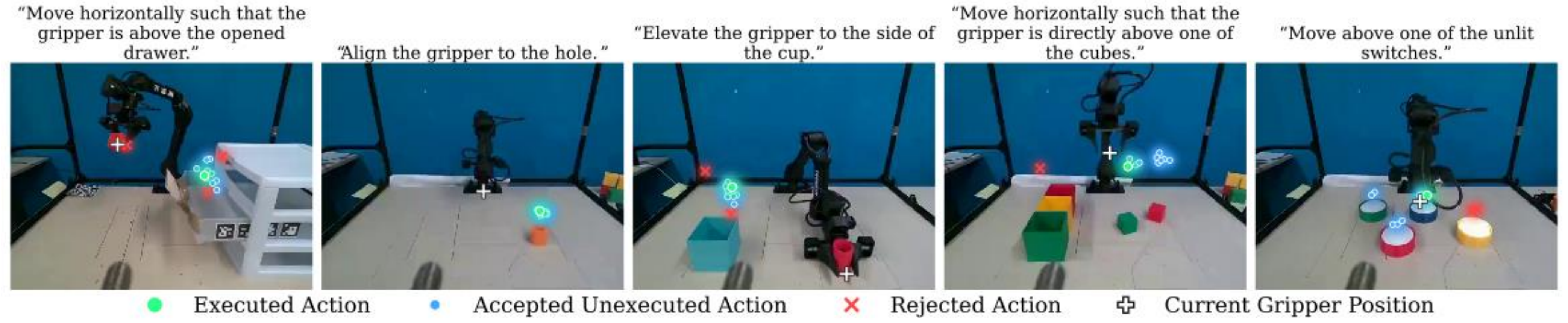
- Recorded states are reconstructed into interactive 3D scenes, where remote contributors specify alternative goal poses.
- Crowdsourced annotations can proceed **asynchronously and in parallel with physical data collection**, without requiring additional access to robot hardware.



(a) Recorded Keyframes (b) Annotation UI

47 participants. 78.7% with no prior robot-arm experience.
7–12 s median submission time.
Submissions validated for task consistency and physical coherence.

Crowdsourced Actions: Diversity and Validation

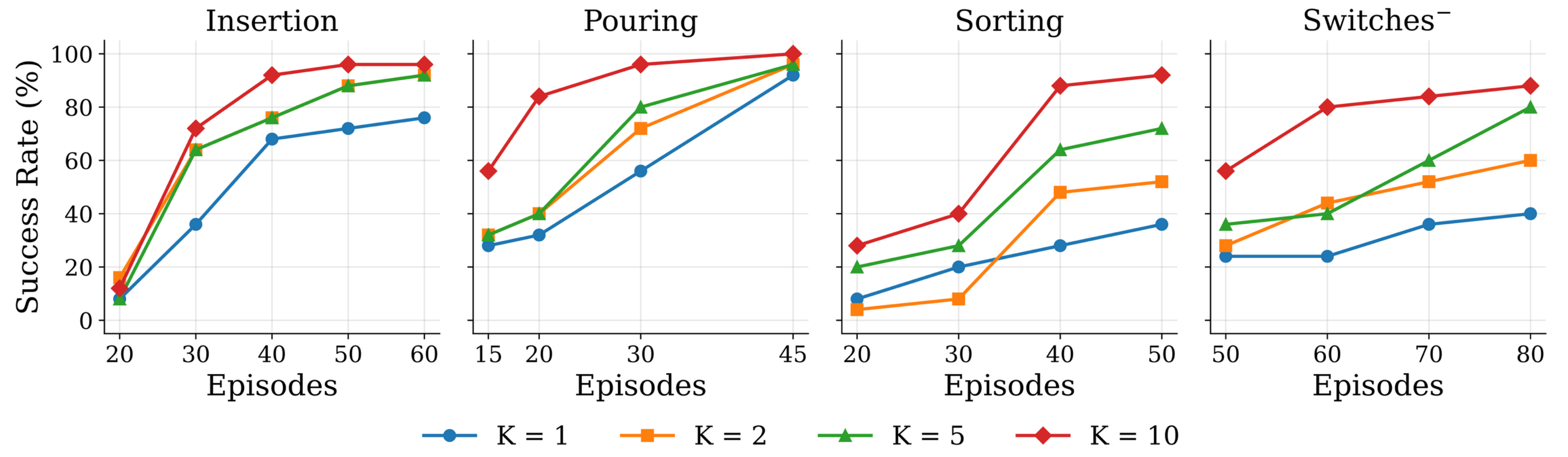

"Move horizontally such that the gripper is above the opened drawer."
"Align the gripper to the hole."
"Elevate the gripper to the side of the cup."
"Move horizontally such that the gripper is directly above one of the cubes."
"Move above one of the unlit switches."
● Executed Action
● Accepted Unexecuted Action
✕ Rejected Action
✜ Current Gripper Position

More Actions per State, Better Policies

Keypose Policies

+35.6 pp

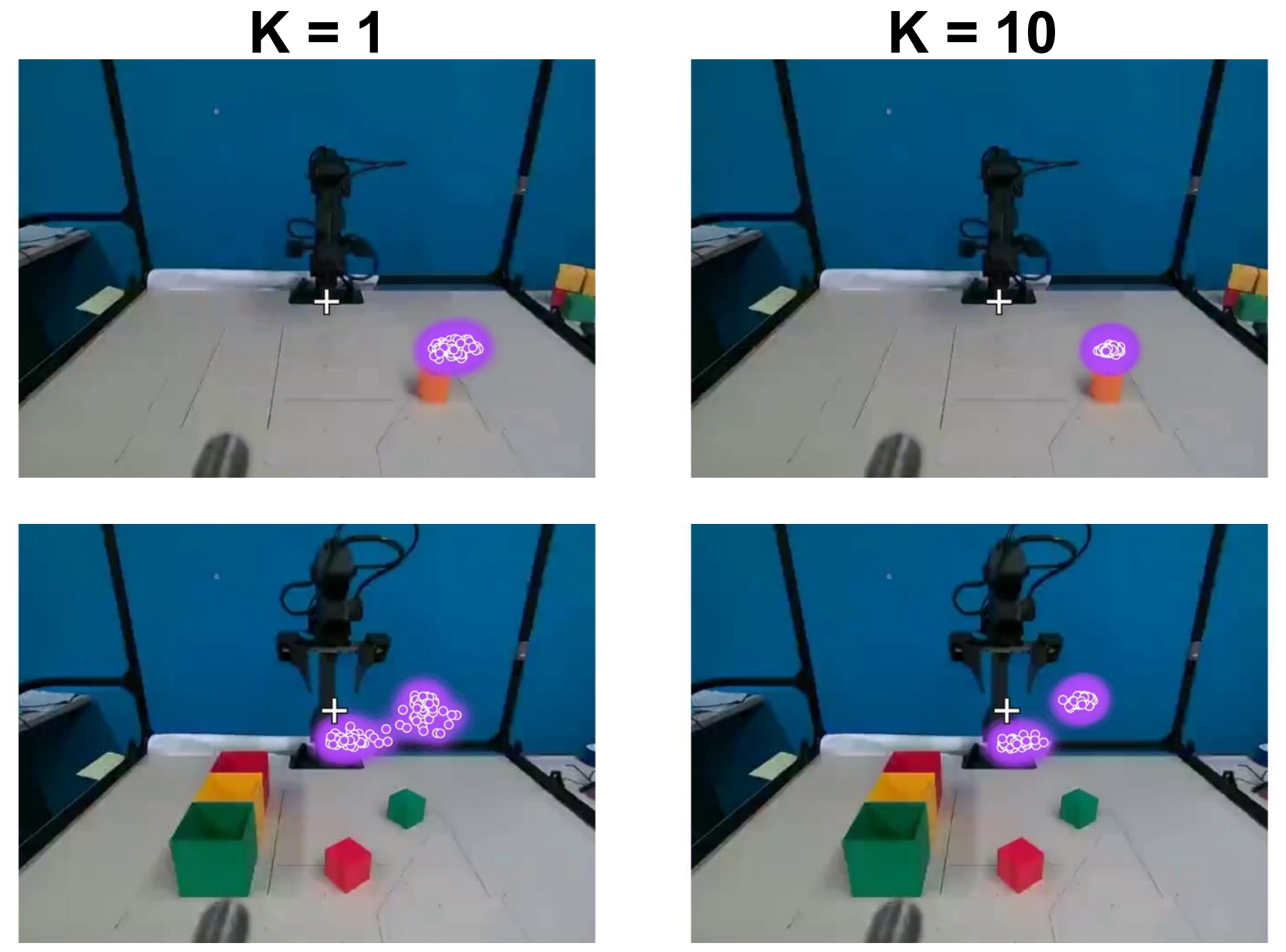
38.7% → 74.2%
Average success: K = 1 → 10



Higher success with fewer physical demonstrations.

Qualitative: Wider MAPS Sharpens Policy Predictions

Held-out keyframes: K = 1 versus K = 10



Insertion
Predictions concentrate more accurately around the valid goal pose.

Sorting
Distinct action modes become sharper.

Trajectory Policies

+33.0 pp

Average success: K = 1 → 10

At comparable robot-time budgets, MAPS (K = 10) outperforms additional expert trajectories on all five tasks.

Takeaway: CrowdMAPS decouples robot data crowdsourcing from robot hardware and densifies supervision per visited state, enabling scalable data collection and more effective robot learning.

Project website: https://yilongsong.github.io/CrowdMAPS/; **or scan the QR code on the right.**